MANIPAL
ACADEMY of HIGHER EDUCATION
(Institution of Eminence Deemed to be University)

MANIPAL SCHOOL OF INFORMATION SCIENCES
(A Constituent unit of MAHE, Manipal)

CYE 5153 Linux OS and Scripting – 2_Lab

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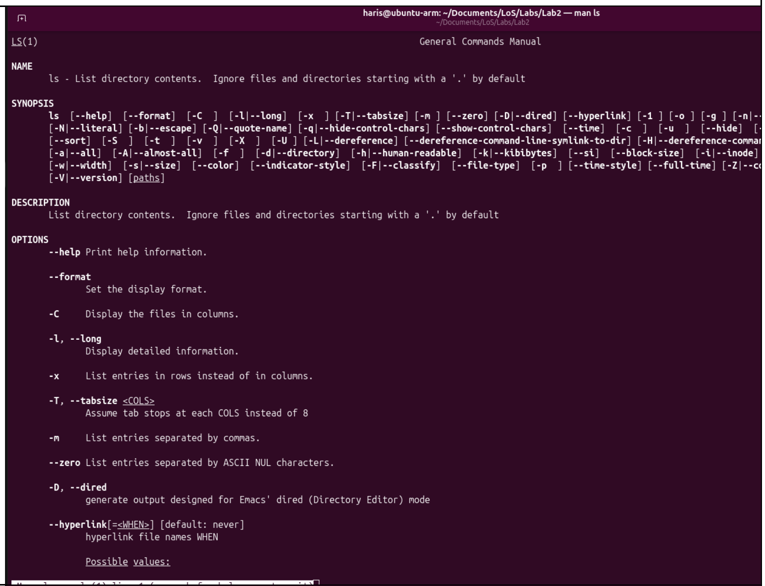
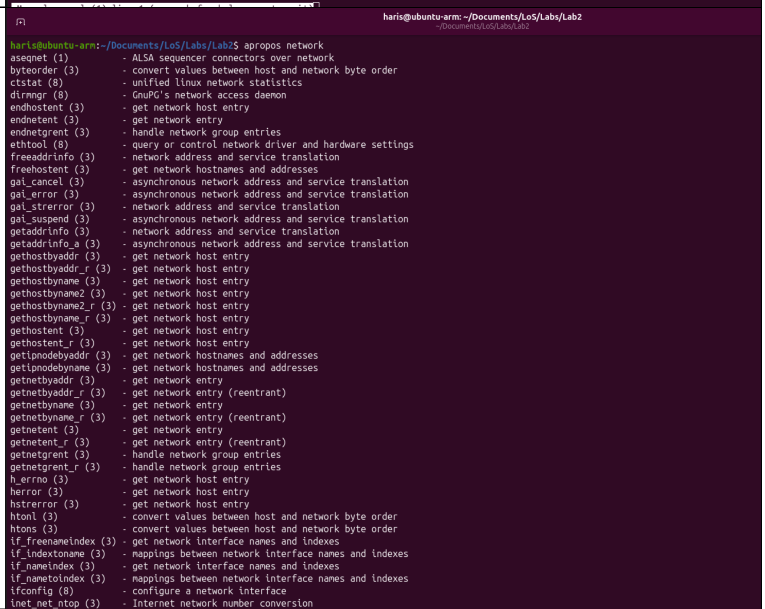
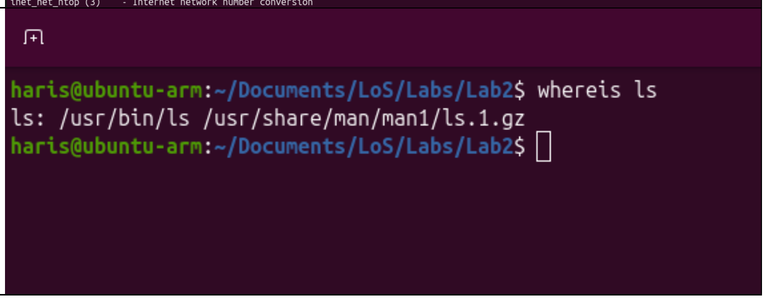
1. Getting Help

<table>
<tr><th>Task</th><th>Command Name</th><th>Syntax</th><th>Example</th><th>Screenshots</th></tr>
<tr><td>To get manual page for the known command</td><td>man</td><td>man [options] [section] command</td><td>man 1 ls</td><td>
</td></tr>
<tr><td>To get manual page for the unknown command</td><td>apropos</td><td>apropos keyword</td><td>apropos network</td><td>
</td></tr>
<tr><td>To know the source file binary</td><td>whereis</td><td>whereis [options] command</td><td>whereis ls</td><td>
</td></tr>
</table>

<table>
<tr><td>To know the path of the command</td><td>which</td><td>which command</td><td>which ls</td><td>haris@ubuntu-arm:~/Documents/LoS/Labs/Lab2$ which ls
/usr/bin/ls
haris@ubuntu-arm:~/Documents/LoS/Labs/Lab2$</td></tr>
<tr><td>To know the command is external or internal</td><td>type</td><td>type command</td><td>type cd</td><td>haris@ubuntu-arm:~/Documents/LoS/Labs/Lab2$ type cd
cd is a shell builtin
haris@ubuntu-arm:~/Documents/LoS/Labs/Lab2$</td></tr>
<tr><td>To get help for the internal command</td><td>help</td><td>help command</td><td>help echo</td><td>haris@ubuntu-arm:~/Documents/LoS/Labs/Lab2$ help cd
cd: cd [-L|[-P [-e]] [-@] [dir]
Change the shell working directory.

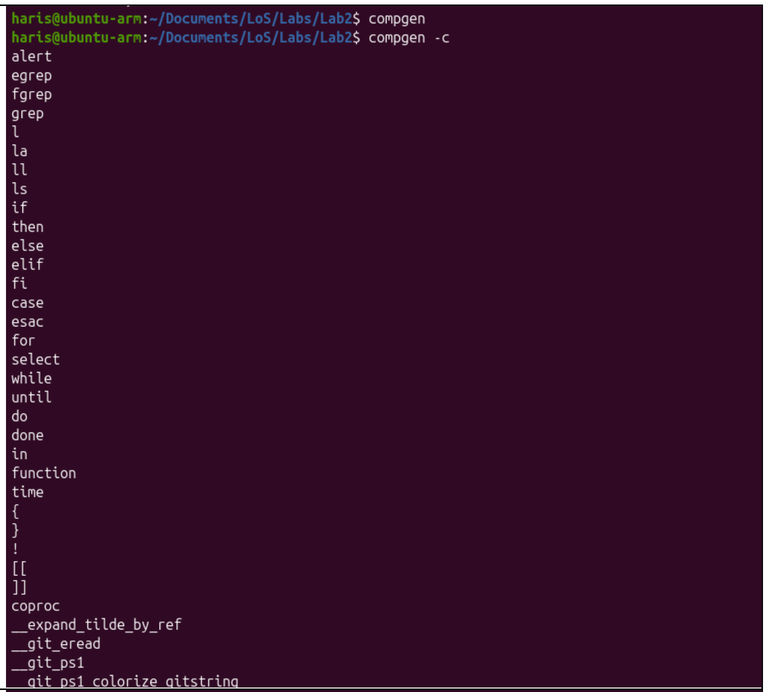
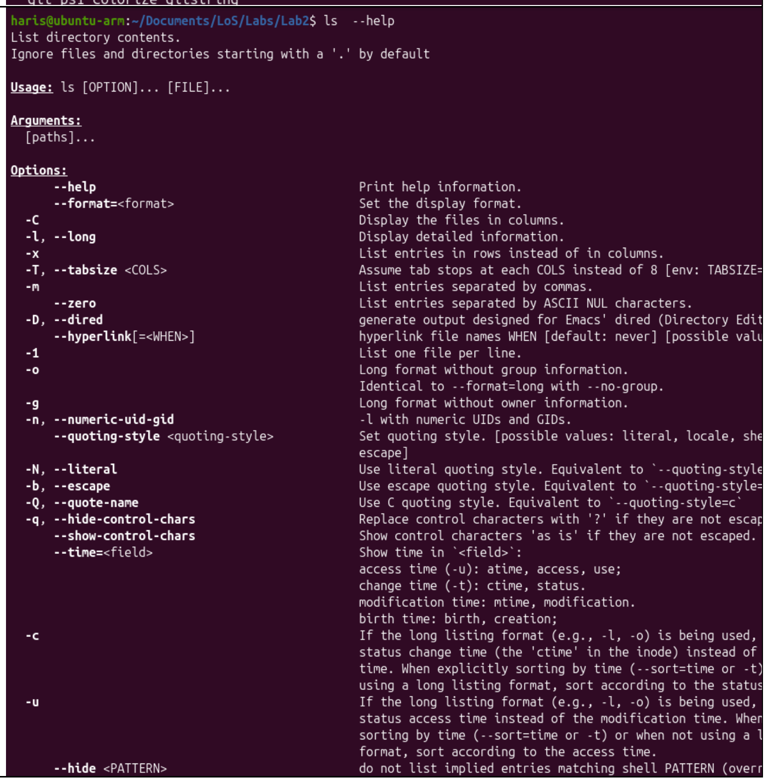
Change the current directory to DIR. The default DIR is the value of the HOME shell variable. If DIR is "-", it is converted to $OLDPWD.

The variable CDPATH defines the search path for the directory containing DIR. Alternative directory names in CDPATH are separated by a colon (:). A null directory name is the same as the current directory. If DIR begins with a slash (/), then CDPATH is not used.

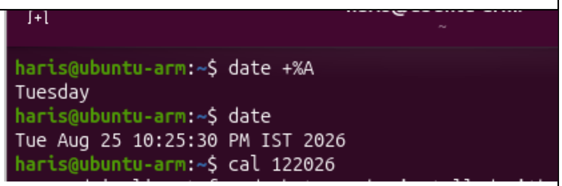
If the directory is not found, and the shell option `cdable_vars' is set, the word is assumed to be a variable name. If that variable has a value, its value is used for DIR.<br><br>Options:<br>-L force symbolic links to be followed: resolve symbolic links in DIR after processing instances of `..'
-P use the physical directory structure without following symbolic links: resolve symbolic links in DIR before processing instances of `..'
-e if the -P option is supplied, and the current working directory cannot be determined successfully, exit with a non-zero status
-@ on systems that support it, present a file with extended attributes as a directory containing the file attributes

The default is to follow symbolic links, as if `-L' were specified. `..' is processed by removing the immediately previous pathname component back to a slash or the beginning of DIR.

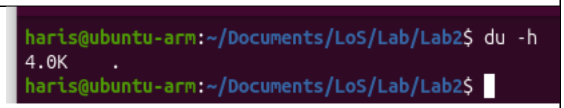
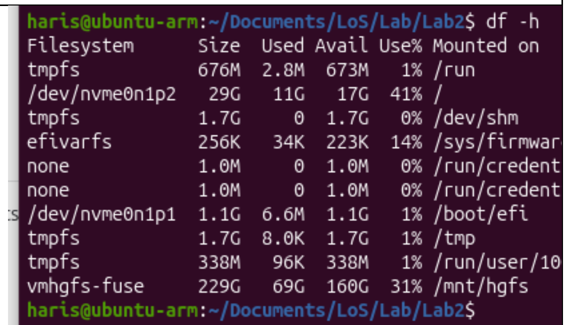
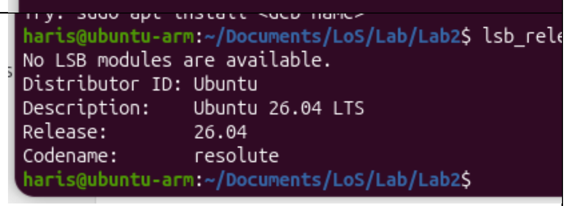
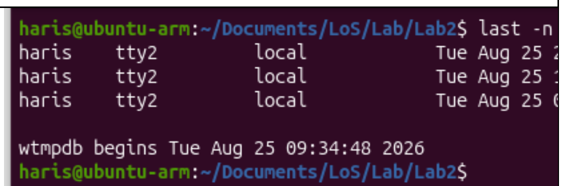
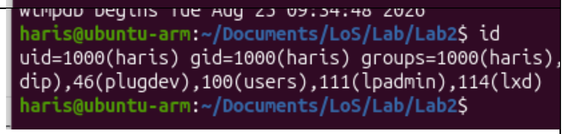
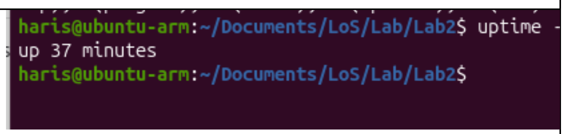
Exit Status:
Returns 0 if the directory is changed, and if $PWD is set successfully when -P is used; non-zero otherwise.
haris@ubuntu-arm:~/Documents/LoS/Labs/Lab2$</td></tr>
</table>

<table>
<tr><td>To list out bash commands</td><td>compgen</td><td>compgen -flag</td><td>compgen</td><td></td><td>
</td></tr>
<tr><td>To know the usage of the command</td><td>help or cheat</td><td>command --help</td><td>ls --help</td><td></td><td>
</td></tr>
</table>

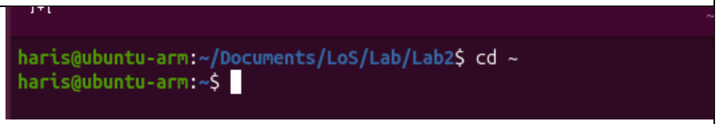
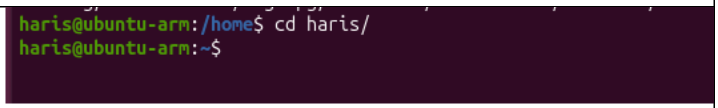
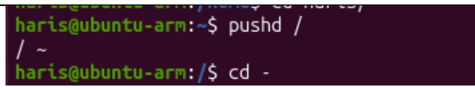
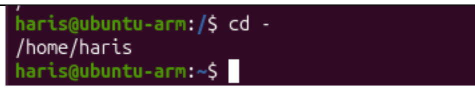
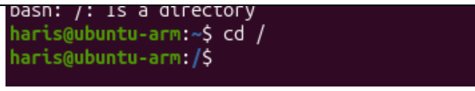
2. Basic Commands

<table>
<tr><th>Task</th><th>Command Name</th><th>Syntax</th><th>Example</th><th>Screenshots</th></tr>
<tr><td>To know today's date</td><td>date</td><td>date [OPTIONS] [+FORMAT]</td><td>date +%A</td><td>
</td></tr>
</table>

To print calendar	cal	cal [OPTIONS] [MONTH] [YEAR]	cal 12 2026	haris@ubuntu-arm:~$ cal 12 2026 December 2026 Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31
To print kernel version	uname	uname [OPTIONS]	uname -r	haris@ubuntu-arm:~/Documents/LoS/Lab/Lab2/Lab2_log_ 7.0.0-30-generic haris@ubuntu-arm:~/Documents/LoS/Lab/Lab2/Lab2_log_
To print default shell	echo $SHELL	echo $SHELL	echo $SHELL	haris@ubuntu-arm:~$ echo $SHELL /bin/bash haris@ubuntu-arm:~$
To print currently logged in user	whoami	whoami	whoami	haris@ubuntu-arm:~$ whoami haris haris@ubuntu-arm:~$
To create shortcut for command	alias	alias shortcut_name='command'	`alias c='clear'`	haris haris@ubuntu-arm:~$ alias c='clear' haris@ubuntu-arm:~$
To delete shortcut	unalias	unalias shortcut_name	unalias clear	haris@ubuntu-arm:~$ unalias clear bash: unalias: clear: not found haris@ubuntu-arm:~$ unalias -a haris@ubuntu-arm:~$
To change the timestamp of the file	touch	touch -d "Time/Date" myfile.txt	touch -t 202608201530.00 myfile.txt	haris@ubuntu-arm:~$ touch -t 202608201530.00 myfil haris@ubuntu-arm:~$ cd haris@ubuntu-arm:~$ ls Desktop Downloads Pictures Templates myfile. Documents Music Public Videos snap haris@ubuntu-arm:~$ stat myfile.txt File: myfile.txt size: 0 Blocks: 0 IO Bloc Device: 259,2 Inode: 568100 Links: 1 Access: (0664/-rw-rw-r--) Uid: (1000/ haris) Access: 2026-08-20 15:30:00.000000000 +0530 Modify: 2026-08-20 15:30:00.000000000 +0530 Change: 2026-08-25 22:31:22.588027792 +0530 Birth: 2026-08-25 22:31:22.585028161 +0530 haris@ubuntu-arm:~$
To clear the screen	clear	clear	clear	
To create	touch	touch [filename]	touch file1.txt	

empty files				
To know disk usage	du	du [OPTIONS] [directory]	du -h	
To know free space in the system	df	df [OPTIONS]	df -h	
To know about the Linux release	lsb_release	lsb_release [OPTIONS]	lsb_release -a	
To show login history	last	last [options] [username] [tty]	last -n 10	
To check the current user privileges	id	id [options] [username]	id	
To check system uptime	uptime	uptime [options]	uptime -p	
To shutdown	shutdown	[sudo] shutdown [options] [time] [message]	sudo shutdown now	
To restart	reboot	[sudo] reboot [options]	sudo reboot	

3. Navigation

Task	Command	Syntax	Screenshots
To navigate home directory	cd ~	cd ~	
To navigate to the parent directory	cd ..	cd ..	
To navigate to the child directory	cd dir	cd directory_name	
Alternate command to cd	pushd and popd	pushd directory_path	
To go back to the previous directory	cd -	cd -	
To go to the root directory	cd /	cd /	

4. File System

Task	Syntax	Command
How to identify the file system	lsblk -f	lsblk

```
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab2$ lsblk -f
NAME        FSTYPE   FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
loop0       squashfs 4.0                                                    0   100% /snap/core24/1588
loop1       squashfs 4.0                                                    0   100% /snap/bare/5
loop2       squashfs 4.0                                                    0   100% /snap/core24/1644
loop3       squashfs 4.0                                                    0   100% /snap/desktop-security-center/149
loop4       squashfs 4.0                                                    0   100% /snap/desktop-security-center/152
loop5       squashfs 4.0                                                    0   100% /snap/firefox/8105
loop6       squashfs 4.0                                                    0   100% /snap/gnome-46-2404/154
loop7       squashfs 4.0                                                    0   100% /snap/mesa-2404/1166
loop8       squashfs 4.0                                                    0   100% /snap/prompting-client/203
loop9       squashfs 4.0                                                    0   100% /snap/prompting-client/221
loop10      squashfs 4.0                                                    0   100% /snap/gtk-common-themes/1535
loop11      squashfs 4.0                                                    0   100% /snap/snap-store/1368
loop12      squashfs 4.0                                                    0   100% /snap/snap-store/1391
loop13      squashfs 4.0                                                    0   100% /snap/snapd/26869
loop14      squashfs 4.0                                                    0   100% /snap/snapd-desktop-integration/363
nvme0n1
├─nvme0n1p1 vfat     FAT32       0020-912D                                 1G     1% /boot/efi
└─nvme0n1p2 ext4     1.0         3f976e8c-039d-439e-af77-9a6ff16d55db     16G    38% /
haris@ubuntu-arm:~/Documents/LoS/Lab/Lab2$
```

5. Globbing

Create a dedicated practice directory containing sample files such as:
index.html, login.html, admin.htm, report.txt, access.log, error.log, user1.txt, user2.log, file123, backup1.html, backup2.txt
Use **globbing patterns wherever possible**, rather than specifying filenames individually.

a. List all files ending in .html.

   *ls *.html*

b. List all files **other than .html files** using extended globbing.

   ls !(.html)*

c. List filenames containing at least one numeric digit.

   *ls *[0-9]**

d. List files whose names begin with user followed by a single digit.

   *ls user?.**

e. List files ending in either .txt or .log using a single **extended glob pattern**.

   *ls *.@(txt|log)*

f. List files that do **not** end in .txt or .log.

   ls !(.txt|*.log)*

g. List files beginning with either access or error using extended globbing.

   ls @(access|error).log

h. List .html files whose filenames contain a numeric digit.

   *ls *[0-9]*.html*

6. Extended Globbing

a. Assume a directory contains:

access.log, error.log, auth.log, auth.log.1, auth.log.2, system.log, notes.txt
Write an extended glob pattern that selects only access.log, error.log, and auth.log.

   ls @(access|error|auth).log

Select all files except notes.txt.

ls !(notes.txt)

Select files named either auth.log.1 or auth.log.2 using pattern matching rather than writing both complete filenames separately.

ls auth.log.@(1|2)

b. Given the files:

malware1.log, malware2.log, malwareA.log, malware_report.txt, normal.log
construct a glob pattern that identifies .log files whose names start with malware and contain a numeric character.

ls malware[0-9]*.log*

c. Given:

failed_login.log, successful_login.log, login_backup.log, login.txt, firewall.log
use extended globbing to select .log files containing either failed_login or successful_login.

ls @(failed_login| successful_login).log

Construct a pattern that lists all .log files **except** firewall.log.

ls !("firewall.log")

7. Filters

a. Display the first **10 entries** of login.log.
head -10 login.log

b. Display the first **5 entries** of access.log.

head -5 access.log

c. Display the first **15 security events** from security.log.

head -15 security.log

d. Display the first **20 events** recorded in incident.log.

head -20 inceident.log

e. Display only the **first entry** in login.log.

head -1 login.log

f. Display the first **25 web requests** recorded in access.log.

head -25 access.log

g. Display the last **10 entries** of login.log.
tail -10 login.log

h. Display the last **5 entries** of access.log.

tail -5 access.log

i. Display the **20 most recent security events** from security.log.

tail -20 security.log

j. Display the **25 most recent events** from incident.log.

tail -25 incident.log

k. Display only the **last entry** in security.log.

tail -1 security.log

l. Display the last **50 events** recorded in incident.log.

tail -50 incident.log

m. Display the contents of login.log starting from **line 41 until the end of the file**.

tail -n +41 login.log

n. Display the contents of security.log starting from **line 101 until the end of the file**.

tail -n +101 security.log

o. Find the total number of lines in login.log.

wc -l login.log

p. Find the total number of lines in access.log.

wc -l access.log

q. Find the total number of security events in security.log.
wc -l security.log

r. Determine the total number of events recorded in incident.log.

wc -l incident.log

s. Count the number of words in login.log.

wc -w login.log

t. Count the number of bytes in security.log.

wc -c security.log

u. Display the **line, word, and byte counts** of access.log.

wc access.log

v. Find the line counts of all four .log files using a single wc command.

wc -l login.log access.log security.log incident.log

w. Display the first **20 entries** of login.log and, from those entries, display only the **last 5**.
head -20 login.log | tail -5

x. Display the last **20 entries** of login.log and, from those entries, display only the **first 5**.

tail -20 login.log | head -5

y. Display the first **30 entries** of access.log and then display only the **last 10 entries** from that output.

head -30 access.log | tail -10

z. Display the last **30 entries** of access.log and then display only the **first 10 entries**.

aa. Display the first **50 events** from security.log and then display the **last 10 events** from those 50.

bb. Display the last **50 events** from security.log and then display the **first 10 events** from those 50.

cc. Display entries from **line 11 to line 20** of login.log.

dd. Display entries from **line 21 to line 30** of login.log.

ee. Display entries from **line 11 to line 20** of access.log.

ff. Display entries from **line 31 to line 40** of access.log.

gg. Display security events from **line 21 to line 30** of security.log.

hh. Display security events from **line 51 to line 60** of security.log.

ii. Display incident events from **line 101 to line 110** of incident.log.

sed -n '101,110p' incident.log

jj. Display incident events from **line 201 to line 220** of incident.log.

kk. Display incident events from **line 491 to line 500**.

ll. Take the first **20 entries** of login.log and count how many lines are displayed.
head -20 login.log | wc -l

mm. Take the last **15 entries** of access.log and count how many lines are displayed.

tail -16 access.log | wc -l

nn. Take the first **50 entries** of security.log and count the number of words in those entries.

oo. Take the last **25 entries** of security.log and count the number of lines.

pp. Extract lines **21–30** from login.log and count the number of extracted lines.

sed -e '21,30p' login.log | wc -l

qq. Extract lines **51–60** from access.log and count the number of extracted lines.

sed -n '51,60p' login.log | wc -l

rr. Extract events **101–150** from incident.log and count how many events are displayed.

ss. Display the first **100 events** from incident.log, take the last **20 events** from that output, and count the number of lines.

head -100 incident.log | tail -20 |wc -l